

Where to pick this up

State as of 2026-08-21. Read this first if resuming.

What is here

file	what
MODEL.md	the report — 14 sections, 9 figures. Also as <code>MODEL.html</code> .
REVIEW-LOG.md	errors found and fixed, who caught each, which checks can fail, why it took as long as it did. Also HTML.
<code>NEXT.md</code>	this file
<code>figs/</code>	the nine figures
<code>*.R</code>	~60 scripts; the report's §13 table says which produces what

Lives in `penm/dev/explorations/scf/enm/`. `dev/` is tracked in git and `.Rbuildignore`, so this is version-controlled but stays out of the package build. `.rds` files are gitignored here — they are regenerable (`catalog.rds` in 3.4 s by `run_catalog.R`).

Scripts locate the package by walking up to `DESCRIPTION`, so the folder can be moved without breaking them.

Run `Rscript validate.R` (8 checks), `Rscript derivation.R` (11 assertions), `Rscript test_identities.R` (4) to confirm the environment still works. Scripts that load `penm` take a minute; `collect_numbers.R` and the trajectory ones take several.

The state of the argument, in one paragraph

Refitting the network to the mutant structure erases the strain, so the energy must be saved in a scalar offset (§§2–3). That works for scanning (§7). It does *not* work for trajectories: re-referencing at every step makes the current state the zero of energy, so every move is uphill and the walk is absorbing (§6, with a projector proof). Julian's catalogue fixes it — reference everything to the founder, and moves become differences of catalogued quantities, so the energy has a fixed floor, the state space is finite, and trajectories reach a stationary state that samples $e^{-\nu V}$ (§8). Frustration is created by every mutation and changes the Hessian by 0.6–2.7 %, but rotates soft modes by < 0.2 % (§9).

What I would do next, in order

1. Give the catalogue states meaning (the obvious next step)

Right now a “state” is a Gaussian draw of δl with no identity. The catalogue structure accommodates amino acids directly: replace *state m of site j* by *residue type a at site j* and nothing else changes. Julian's 2017 appendix lists this as strategy 2 — l_{ij} depending only on the identities of the residues at i and j , giving 210 independent parameters fitted against the $3N - 6$ equations. That turns the model from a random-perturbation toy into something that could be compared with real substitution data.

Concretely: build the catalogue with 20 states per site, one per amino acid, with δl_{ij} drawn from a distribution conditioned on the residue pair. The trajectory machinery in `penm_catalog.R` needs no change.

2. Fix penm's `sclfenm` (§11) — one line of iteration

`get_mutant_site_sclfenm()` takes one linear step, rebuilds the graph around the result, and never re-relaxes against the new graph. The mutant sits at $|F| = 0.71$ on its own network, 33× worse than the LFENM mutant, and the effect is driven entirely by *lost* contacts (gained ones arrive relaxed and cost nothing).

`relax_to_min()` in `frustration.R` does the iteration in ~5 Newton steps. I did not touch the package. Before changing it, note that the `sclfenm` fixtures are frozen and the tests skip — see `penm/CLAUDE.md`, which says to stop and ask before touching `sclfenm`.

3. Decide whether the catalogue needs the cross-term correction

Background-independence degrades: 9 % $\rightarrow$ 48 % (energy), 0.7 % $\rightarrow$ 17 % (structure), between 5 and 40 substitutions. The energy error is *entirely* the cross term (correlation 0.9999), and the cross term is computable from the current state's strain. Correcting for it is the cheapest way to extend the catalogue's useful range; the alternative is rebuilding the catalogue every few tens of substitutions.

4. Port §§1–7 onto penm's functions

The report's first seven sections use a reimplementations (`enm_core.R`, `sclfenm.R`) written before I read `R/penm.R`; §§8–11 use the package. They agree to machine precision (`test_two_impls.R`), but two code paths for one piece of physics is a defect. Note the nuance in `REVIEW-LOG.md` §4: independence is *correct* for `sclfenm` and `frustration` — those are what is under investigation — and waste for the settled parts.

Open questions this exploration did not settle

- Per-move accuracy of the catalogue. Aggregate energy tracks to 2 %, but on a 30-substitution background 1 move in 12 had the *wrong sign*. Fine for ensemble properties; not fine for a study of individual substitutions.
- Site-additivity at larger σ . Measured within 2 % at $\sigma = 0.3$ on 2acy. Untested elsewhere.
- Whether frustration matters for anything penm computes. Measured effect is small (§9) but the `frustrated = TRUE` branch has still never been exercised in a trajectory.
- Path-independence under the rebuild on a real protein. The 6×10^{-4} figure in §4 is from a synthetic 40-node helix, the one result here not measured on 2acy.
- SC-LFENM's irreversibility (`sc_lfenm.md`'s 2022 #todo). The catalogue makes the *bookkeeping* reversible, which is not the same as showing the underlying model is.

Things to be careful of (they cost me time)

1. `get_eij()` is not refreshed by `get_mutant_site_lfenm()`. LFENM never rebuilds `K`, so penm has no reason to. If you build a Hessian from penm's stored `eij` after an LFENM mutation, you mix wild-type directions with mutant distances. Recompute from `xyz`.
2. Spectra must be taken at a stationary point. The LFENM structure is the linear-response estimate, not a minimum. A Hessian evaluated there is not singular on rotations and produces three spurious modes $\sim 4 \times 10^{-4}$, which put *TS* out by 8.6. `relax_to_min()` first.
3. β is Boltzmann's $1/kT$; selection strength is ν . Conflating them silently broke a figure once.
4. The report uses `mut_sd_min = 1`, not penm's default of 2. Every contact of the mutated site is perturbed, backbone neighbours included.
5. penm blocks `frustrated = TRUE` in `set_enm()`. The frustrated Hessian here is built directly in `frustration.R`, bypassing the package.